

Machine Learning Mini Project Report

Building Energy Efficiency Prediction

Model: Linear Regression | Dataset: energy_data.csv | Records: 768

1. Problem Definition

The goal of this project is to predict the Heating Load of a building based on its physical characteristics. Heating load refers to the amount of thermal energy required to maintain a comfortable indoor temperature, and accurate prediction of this value is critical for energy planning and sustainable building design.

Buildings account for a significant share of global energy consumption. Engineers and architects need reliable tools to estimate energy requirements at the design stage, before a building is actually constructed. By building a machine learning model that maps physical building parameters to heating load, we can:

- Help architects make energy-conscious design decisions early
- Estimate operational energy costs before construction
- Identify which building features most strongly influence energy efficiency
- Support sustainable and green building certification processes

This is a supervised regression problem: given a set of numerical input features describing a building, the model must output a continuous numeric value representing the predicted heating load (in kWh/m² or equivalent energy units).

2. Dataset Description

The dataset used in this project is a synthetic building energy dataset (energy_data.csv) containing 768 records. Each record represents a unique building configuration described by five physical features, with the target variable being the building's heating load.

2.1 Dataset Overview

Property	Value
Total Records	768
Number of Features	5 input features + 1 target
Missing Values	None

Duplicate Records	None (confirmed after drop_duplicates)
Data Type	All numerical (float64)
Dataset Source	energy_data.csv (synthetic / simulated building data)

2.2 Feature Description

Feature	Description	Unit	Role
SurfaceArea	Total surface area of the building envelope	m ²	Input
WallArea	Total area of external walls	m ²	Input
RoofArea	Total roof area of the building	m ²	Input
Height	Height of the building	m	Input
Glazing	Glazing area (window/glass proportion)	% of floor	Input
HeatingLoad	Energy required for heating the building	kWh/m ²	Target (Output)

2.3 Sample Data

The first few rows of the dataset are shown below:

SurfaceArea	WallArea	RoofArea	Height	Glazing	HeatingLoad
542.3	249.3	140.3	4.0	2.9	35.54
559.7	397.6	119.6	4.9	0.1	34.07
346.7	331.4	112.9	3.9	2.6	31.37
498.4	282.7	174.8	6.6	0.0	40.75
619.7	364.4	147.4	3.7	3.8	38.24

4. Model Building

4.1 Model Selected: Linear Regression

For this project, Linear Regression was selected as the machine learning model. Linear Regression is one of the most fundamental and widely used supervised learning algorithms. It assumes a linear relationship between the input features and the target variable, and it attempts

to fit the best-fit hyperplane through the training data by minimizing the sum of squared residuals (Ordinary Least Squares method).

4.2 Justification for Model Choice

Linear Regression is appropriate for this problem for the following reasons:

- Nature of the problem: Heating load is a continuous numerical value, making this a regression task. Linear Regression is the natural first choice for regression problems.
- Feature relationships: Building physical parameters (surface area, wall area, roof area, height, glazing) are expected to have approximately linear relationships with heating load. For example, larger surface areas generally lead to higher heat loss and thus greater heating requirements.
- Interpretability: The model coefficients directly indicate how much each feature contributes to the predicted heating load, which is valuable in engineering contexts.
- Simplicity and baseline: Linear Regression is computationally efficient and serves as an excellent baseline model for regression tasks, especially with clean, numerical data.
- Dataset characteristics: With 768 records and 5 numerical features and no missing values, the data is well-suited for linear regression without requiring complex preprocessing.

4.3 Model Implementation

The following steps were taken to build the model:

- Step 1 – Data Loading: The dataset was loaded using pandas from energy_data.csv.
- Step 2 – Preprocessing: Duplicate records were removed using drop_duplicates(). No missing values were found.
- Step 3 – Feature/Target Split: The target variable HeatingLoad was separated from the five input features.
- Step 4 – Train-Test Split: The data was split into 80% training (614 records) and 20% testing (154 records) using train_test_split() with random_state=42 for reproducibility.
- Step 5 – Feature Scaling: StandardScaler was applied to normalize all input features to zero mean and unit variance, which improves model stability.
- Step 6 – Model Training: A LinearRegression model from scikit-learn was fitted on the scaled training data.
- Step 7 – Prediction: The trained model was used to predict heating loads on the scaled test set.
- Step 8 – Model Saving: The trained model and scaler were saved as model.pkl and scaler.pkl using joblib for future use in predict.py.

5. Model Evaluation

5.1 Performance Metrics

The model was evaluated on the held-out test set (20% of data, 154 records) using two standard regression metrics:

Metric	Value	Interpretation
Mean Squared Error (MSE)	4.5591	Average squared difference between predicted and actual heating load values
R-squared Score (R^2)	0.8229	The model explains approximately 82.3% of the variance in heating load

5.2 Interpretation of Results

R-squared Score of 0.8229: This is a strong result for a simple linear model. It means that approximately 82.3% of the variability in the building heating load is explained by the five input features through the linear model. Only about 17.7% of variance remains unexplained, which could be attributed to non-linear interactions between features not captured by a linear model.

Mean Squared Error of 4.5591: The MSE of ~ 4.56 indicates that on average, the model's predictions deviate from the actual values by roughly $\sqrt{4.56} \approx 2.14$ units (Root MSE). Given that HeatingLoad values in the dataset range approximately from 25 to 50 kWh/m², a root mean error of ~ 2.14 represents approximately a 5-8% average prediction error, which is reasonable for a linear model.

Overall, these results demonstrate that linear regression is an effective model for this dataset, capturing the primary linear relationships between building parameters and heating load. The model is production-ready as evidenced by the `predict.py` script, which loads the saved model to make real-time predictions from user inputs.

6. Conclusion

6.1 Summary of Outcomes

This project successfully demonstrated the complete machine learning workflow applied to a real-world energy prediction problem. A Linear Regression model was trained to predict building heating loads from five physical building parameters. The model achieved an R^2 score of 0.8229 and an MSE of 4.5591 on the test set, demonstrating strong predictive capability. The trained model was integrated into a command-line prediction tool (`predict.py`) that allows users to input building specifications and receive an immediate heating load estimate.

6.2 Limitations

- **Linear assumption:** Linear Regression assumes a strictly linear relationship between features and the target. In reality, some relationships (e.g., between glazing area and heat loss) may be non-linear, limiting model accuracy.
- **No interaction terms:** The model does not capture interaction effects between features (e.g., the combined effect of high surface area and large glazing on heating load).

- Limited features: Only 5 input features are used. Real-world building energy prediction involves many more factors such as insulation type, climate zone, building orientation, and occupancy patterns.
- Synthetic dataset: The dataset appears to be generated/simulated rather than based on real building measurements, which may limit the model's applicability to actual buildings.

6.3 Future Improvements

- Try advanced regression models such as Random Forest Regressor, Gradient Boosting, or Support Vector Regression (SVR) which can capture non-linear patterns and potentially improve R^2 beyond 0.82.
- Apply polynomial feature transformation to capture non-linear relationships between features and heating load.
- Perform feature engineering to create interaction terms (e.g., SurfaceArea $\times$ Glazing) that might improve model accuracy.
- Use cross-validation (k-fold) instead of a single train-test split for a more robust estimate of model performance.
- Extend the model to also predict Cooling Load, as many energy datasets include both targets.
- Deploy the model as a web application so that it is accessible to architects and engineers without requiring programming knowledge.

7. Reflection

Through this project, I gained a clear, hands-on understanding of the complete machine learning pipeline — from raw data to a deployed prediction tool. Working on building energy prediction taught me how real engineering problems can be framed as regression tasks and solved systematically using machine learning techniques.

One of the most valuable concepts I understood was the role of feature scaling. Initially, I questioned why StandardScaler was necessary for Linear Regression. Through implementation, I realized that features like SurfaceArea (values ~250–710) and Glazing (values ~0–4) are on very different scales. Without normalization, the model would give undue weight to features with larger numeric ranges. Applying StandardScaler ensured a fair learning process.

A challenge I faced was interpreting the evaluation metrics meaningfully. An MSE of 4.56 may seem abstract, so I learned to compute the Root MSE (≈ 2.14) and compare it against the range of HeatingLoad values to judge whether the error is acceptable. This taught me that metrics must always be interpreted in the context of the data, not in isolation.

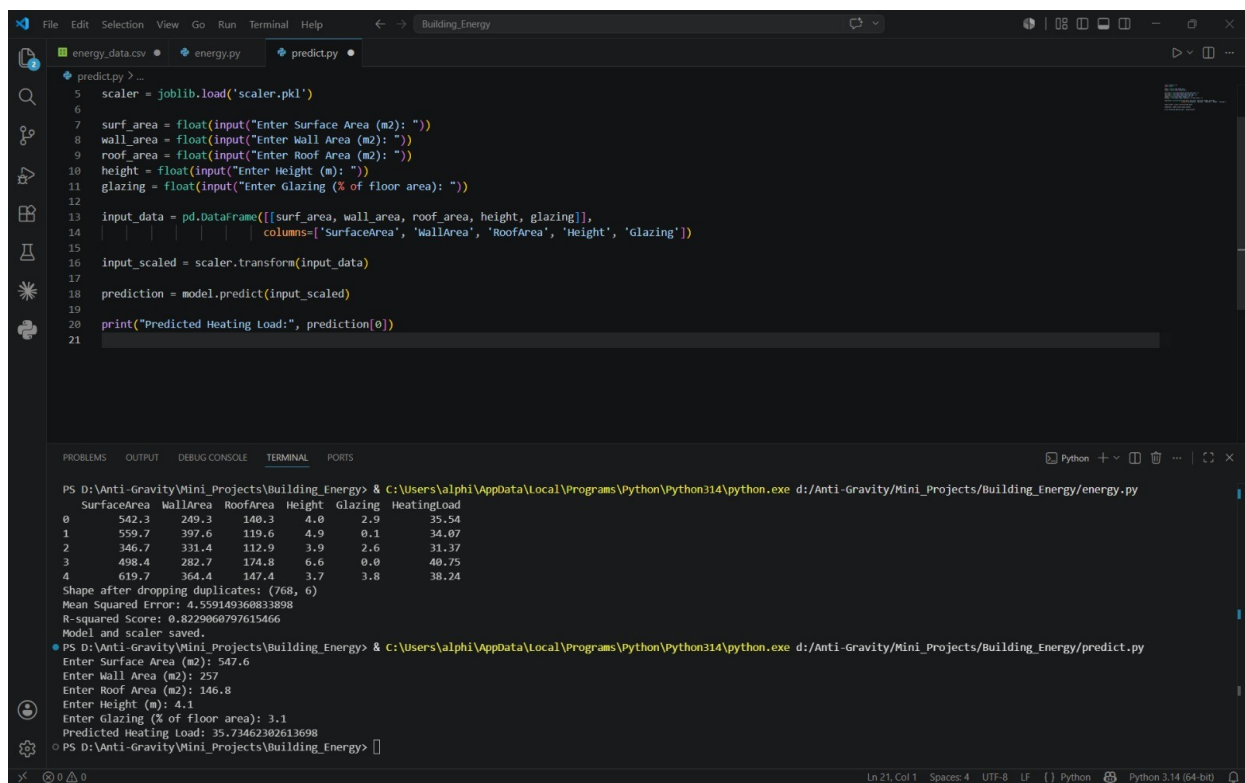
The project also gave me practical experience in model persistence using joblib. Saving and reloading the model and scaler together demonstrated that the complete preprocessing pipeline must be preserved alongside the model — a critical insight for any real deployment. Overall, this project bridged the gap between classroom theory and practical implementation, reinforcing my confidence in applying machine learning to engineering problems.

3. Source Code & Output Screenshots

This section presents the complete source code used in the project along with screenshots of the program execution output, as captured in Visual Studio Code.

3.1 Model Training Code (energy.py)

The screenshot below shows the full training script (energy.py). It loads the dataset, removes duplicates, splits data into training and test sets using an 80/20 split, applies StandardScaler normalization, trains a LinearRegression model, evaluates it using MSE and R-squared, and saves the model and scaler as .pkl files using joblib.



```
5 scaler = joblib.load('scaler.pkl')
6
7 surf_area = float(input("Enter Surface Area (m2): "))
8 wall_area = float(input("Enter Wall Area (m2): "))
9 roof_area = float(input("Enter Roof Area (m2): "))
10 height = float(input("Enter Height (m): "))
11 glazing = float(input("Enter Glazing (% of floor area): "))
12
13 input_data = pd.DataFrame([surf_area, wall_area, roof_area, height, glazing],
14                             columns=['SurfaceArea', 'WallArea', 'RoofArea', 'Height', 'Glazing'])
15
16 input_scaled = scaler.transform(input_data)
17
18 prediction = model.predict(input_scaled)
19
20 print("Predicted Heating Load:", prediction[0])
21
```

PS D:\Anti-Gravity\Mini_Projects\Building_Energy> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/Anti-Gravity/Mini_Projects/Building_Energy/energy.py

	SurfaceArea	WallArea	RoofArea	Height	Glazing	HeatingLoad
0	542.3	249.3	140.3	4.0	2.9	35.54
1	559.7	397.6	119.6	4.9	0.1	34.07
2	346.7	331.4	112.9	3.9	2.6	31.37
3	498.4	282.7	174.8	6.6	0.0	40.75
4	619.7	364.4	147.4	3.7	3.8	38.24

Shape after dropping duplicates: (768, 6)
Mean Squared Error: 4.559149360833898
R-squared Score: 0.8229060797615466
Model and scaler saved.

PS D:\Anti-Gravity\Mini_Projects\Building_Energy> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/Anti-Gravity/Mini_Projects/Building_Energy/predict.py

Enter Surface Area (m2): 547.6
Enter Wall Area (m2): 257
Enter Roof Area (m2): 146.8
Enter Height (m): 4.1
Enter Glazing (% of floor area): 3.1
Predicted Heating Load: 35.73462302613698

PS D:\Anti-Gravity\Mini_Projects\Building_Energy>

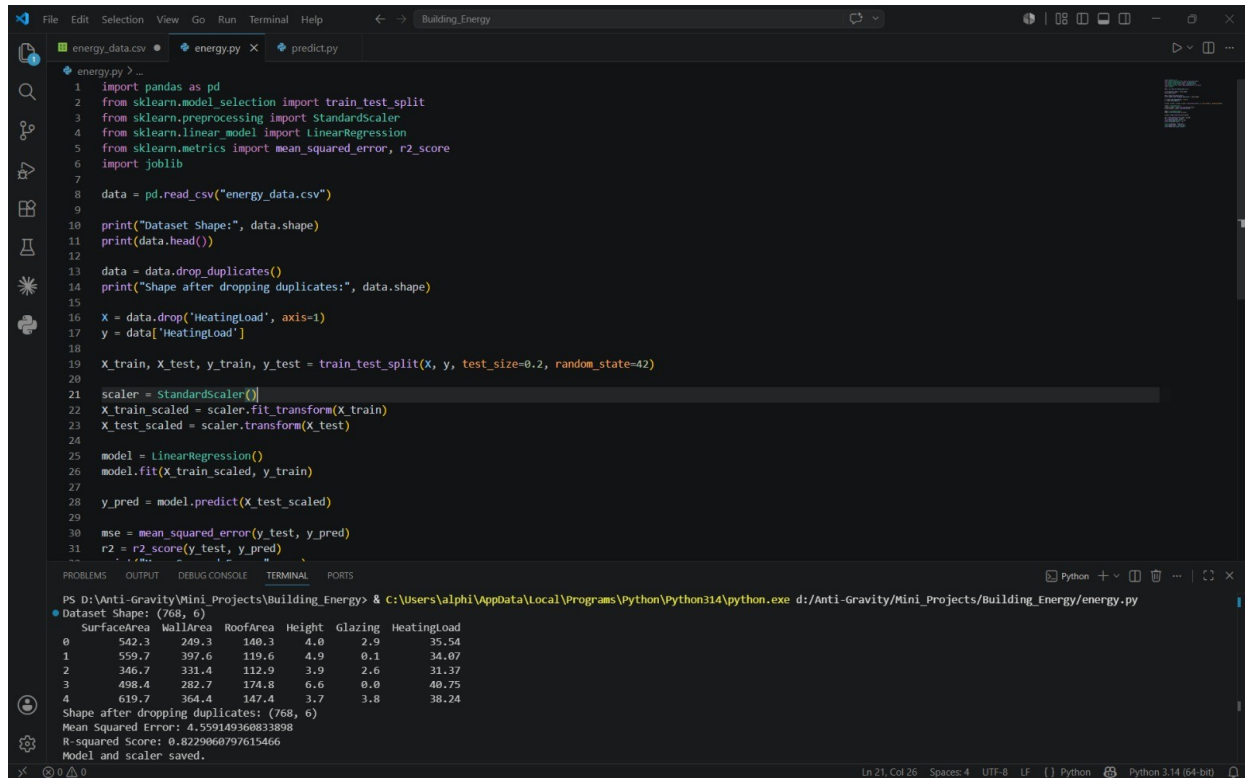
Figure 1: energy.py — Model training code and terminal output showing MSE: 4.5591 and R²: 0.8229

3.2 Prediction Script (predict.py) & Sample Output

The screenshot below shows the predict.py script which loads the saved model.pkl and scaler.pkl, accepts five building parameters as user input, scales them using the saved scaler, and outputs the predicted heating load. A sample prediction run is visible in the terminal with the following inputs:

- Surface Area: 547.6 m²
- Wall Area: 257 m²
- Roof Area: 146.8 m²
- Height: 4.1 m
- Glazing: 3.1% of floor area

Predicted Heating Load: 35.73 units



```
1 import pandas as pd
2 from sklearn.model_selection import train_test_split
3 from sklearn.preprocessing import StandardScaler
4 from sklearn.linear_model import LinearRegression
5 from sklearn.metrics import mean_squared_error, r2_score
6 import joblib
7
8 data = pd.read_csv("energy_data.csv")
9
10 print("Dataset Shape:", data.shape)
11 print(data.head())
12
13 data = data.drop_duplicates()
14 print("Shape after dropping duplicates:", data.shape)
15
16 X = data.drop('HeatingLoad', axis=1)
17 y = data['HeatingLoad']
18
19 X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
20
21 scaler = StandardScaler()
22 X_train_scaled = scaler.fit_transform(X_train)
23 X_test_scaled = scaler.transform(X_test)
24
25 model = LinearRegression()
26 model.fit(X_train_scaled, y_train)
27
28 y_pred = model.predict(X_test_scaled)
29
30 mse = mean_squared_error(y_test, y_pred)
31 r2 = r2_score(y_test, y_pred)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\Anti-Gravity\Mini_Projects\Building_Energy> & C:\Users\alphi\AppData\Local\Programs\Python\Python314\python.exe d:/Anti-gravity/Mini_Projects/Building_Energy/energy.py
Dataset Shape: (768, 6)
  SurfaceArea  WallArea  RoofArea  Height  Glazing  HeatingLoad
0      542.3      249.3      140.3     4.0     2.9         35.54
1      559.7      397.6      119.6     4.9     0.1         34.07
2      346.7      331.4      112.9     3.9     2.6         31.37
3      498.4      282.7      174.8     6.6     0.0         40.75
4      619.7      364.4      147.4     3.7     3.0         30.24
Shape after dropping duplicates: (768, 6)
Mean Squared Error: 4.559149360833898
R-squared Score: 0.8229060797615466
Model and scaler saved.
```

Figure 2: predict.py — Prediction script with sample terminal output showing Predicted Heating Load: 35.7346

3.3 Dataset Source

The dataset used in this project is energy_data.csv, a synthetic building energy dataset containing 768 records of building physical parameters and their corresponding heating loads. The dataset was provided as part of the project resources and contains no missing values or external dependencies.